

ADAPTIVE SCIENTIFIC REASONING · NATURE FOUNDATION MODELS

ASRA-NFM

Learning world dynamics through intervention, explicit mechanism discovery, and planning under uncertainty.

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DEFINITION

Not a predictor wrapper — an adaptive reasoner

A neuro-symbolic architecture that learns world dynamics through intervention, represents discovered mechanisms explicitly, and plans under uncertainty.

- Observes structured environments
- Performs interventions and records before–action–after transitions
- Maintains competing hypotheses
- Selects actions that reduce uncertainty or advance an inferred goal

CENTRAL RESEARCH QUESTION

Can an agent learn unfamiliar environments efficiently by treating actions as scientific interventions, maintaining competing explanations, and explicitly modeling the mechanisms that produce observed changes?

MOTIVATION

Why hybrid active science?

- Pattern recognition can predict without discovering mechanisms
- Purely symbolic systems break under uncertain perception
- Passive learning is inefficient in novel interactive worlds
- Need: action meaning, goals, mechanisms, planning with incomplete knowledge

NEURAL

SYMBOLIC

INTERVENTION

FOUNDATIONAL UNIT

Before → Action → After

$$T_t = (s_t, a_t, s_{t+1}) \quad \Delta_t = \text{Diff}(s_t, s_{t+1})$$

- s_t — structured state before the action
- a_t — action / intervention
- s_{t+1} — resulting structured state
- Δ_t — observed effect of the intervention

Evidence for action semantics, rules, mechanisms, invariants, goals, terminals.

ARCHITECTURE

Closed scientific loop

 ASRA-NFM architecture

Neural · symbolic · memory/evidence · planning — with feedback into hypothesis revision.

REASONING ORGANIZATION

Three functional modules

Tabby — Questions

Identifies uncertainty; proposes discriminating interventions.

Grubby — Evidence

Supports, contradictions, invariants, counterexamples from transitions.

Sam — Decisions

Explore, test, or execute a goal-directed plan.

Modules — not a requirement for three separate language models.

REPRESENTATION

Object-centric reasoning

Objects

identity · type · color · shape · position · attributes · history

Relations

adjacent · inside · aligned · blocking · approaching · connected

$$\text{Effect}(a, o_i, r_{ij}, c_t) \rightarrow \Delta o_i$$

Must influence hypotheses, prediction, experiment selection, scoring, and planning.

MECHANISMS

Competing falsifiable hypotheses

$$H_t = \{h_1 \dots h_k\}$$

$$\text{Score}(h) = \text{Acc} + \alpha \text{Cov} - \beta \text{Complexity} - \gamma \text{Contradictions}$$

- H1: ACTION_2 moves every red object
- H2: moves the object adjacent to the agent
- H3: moves the selected object when the path is clear

Interventions distinguish these possibilities.

ACTION SELECTION

Progress and information gain

$$a^* = \operatorname{argmax}_a [IG(a; H_t) + \lambda P(\text{progress} \mid a) - \mu \text{Risk}(a) - \nu \text{Cost}(a)]$$

- Advance the inferred objective
- Reveal which world-model hypothesis is correct

MEMORY

Evidence that accumulates

Episodic

Complete runs

Transition

Before–action–after

Semantic

Concepts and rules

Procedural

Successful sequences

Failure

Contradicted rules

Experiment

Hypotheses, configs, conclusions

SCOPE

From interactive testbeds to science

- Initial: ARC-AGI-style interactive environments
- Novel interactive environments · autonomous ML experimentation
- Scientific discovery · robotics · biological decision systems · physical AI

Competition adapters share one adaptive scientific-reasoning core.

METHODOLOGY

Comparisons and ablations

Baselines

Random · heuristic · neural-only · symbolic-only · hybrid · hybrid + active IG

Ablations

– relations · – hypothesis memory · – neural WM · – active experiment · – failure memory ·
– complexity penalty

Metrics: success, score, efficiency, prediction, mechanism recovery, goal inference, IG/action, calibration, compute, generalization.

RESULTS

Placeholders — no invented numbers

Benchmark [BENCHMARK] · Score [SCORE] · Previous [PREVIOUS_SCORE] · Δ [DELTA]
N [N] · Date [DATE] · Version [VERSION] · Hardware [HARDWARE]

Placeholder charts

Preliminary result; further controlled evaluation and ablation studies are in progress.

HYPOTHESIS

Main research claim

An object-centric neuro-symbolic agent with explicit hypothesis management and active intervention can learn novel mechanisms more accurately and with fewer actions than passive, neural-only, and symbolic-only baselines.

Presented as a hypothesis until validated by controlled evaluation.

CONTRIBUTIONS

Intended contributions

1. Transition-centered learning
2. Separation of action semantics, mechanisms, and goals
3. Object-centric causal representation
4. Competing, falsifiable hypotheses
5. Hybrid neural + symbolic world models
6. Information-seeking interventions
7. Interpretable evidence and decision traces
8. Reusable core for interactive reasoning and scientific experimentation

NEXT

ASRA-NFM

Intervene. Explain. Revise. Plan.

[Project URL] · [GitHub] · [Email]

Poster pack: [private/marketing/theory/posters](#) · [Editable HTML/SVG](#) · [No AGI claims](#)